

Sesi 11

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The Short-Run Trade-Off between Inflation and Unemployment

Pengantar Ekonomi 2

Sumber: Mankiw (Ch 36)

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Inflation and Unemployment

- In the long run, inflation and unemployment are unrelated:
 - The inflation rate depends mainly on growth in the money supply
 - Natural rate of unemployment depends on various features of the labor market
 - Job search, minimum-wage laws, union power, and efficiency wages

In the short run, society faces a trade-off between inflation and unemployment.

Deriving the Phillips Curve

Suppose $P = 100$ this year.

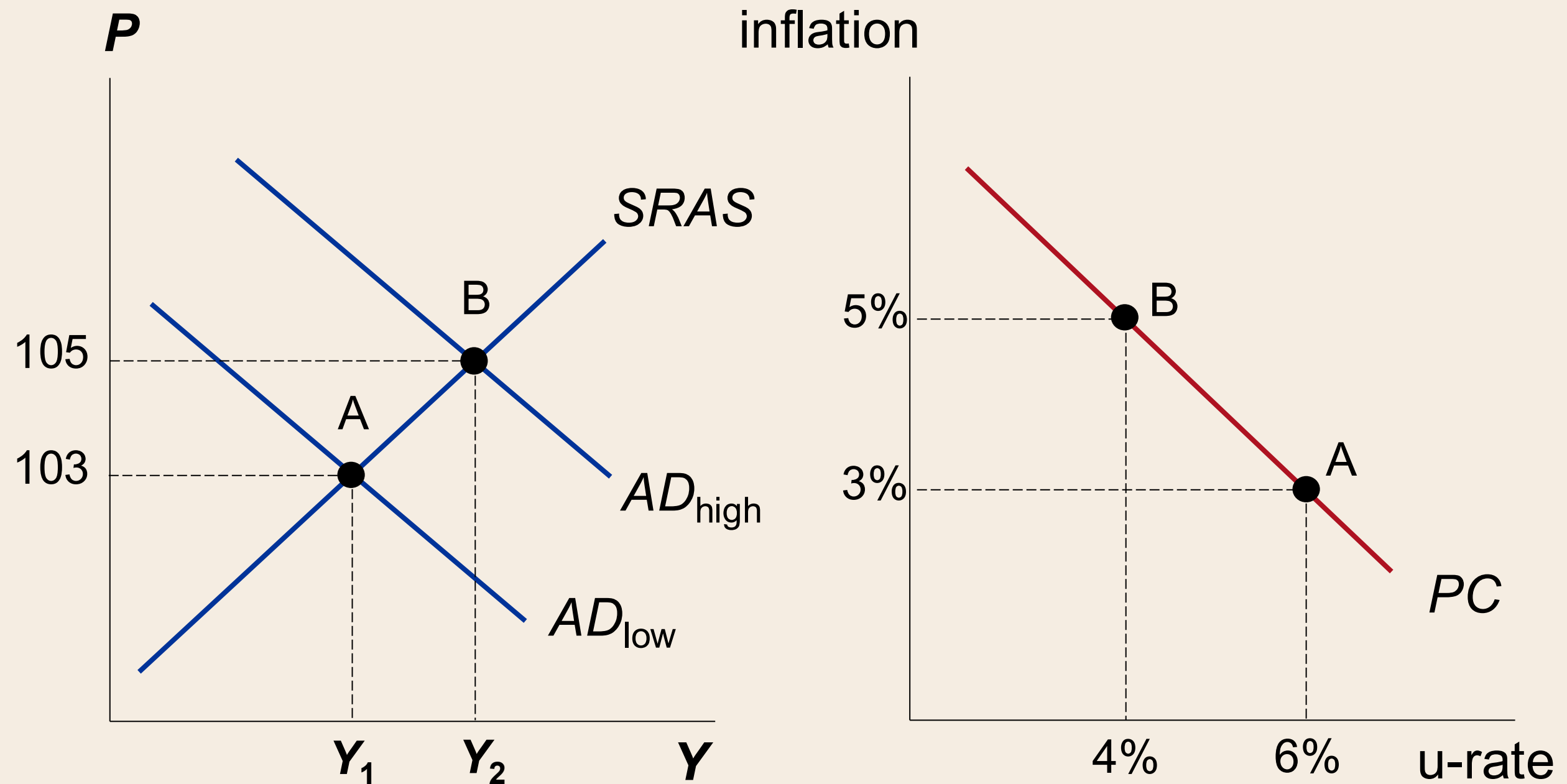
The following graphs show two possible outcomes for next year:

Point A:

Low aggregate demand, small increase in P (i.e., low inflation), low output, high unemployment.

Point B:

High aggregate demand high, big increase in P (i.e., high inflation), high output, low unemployment.



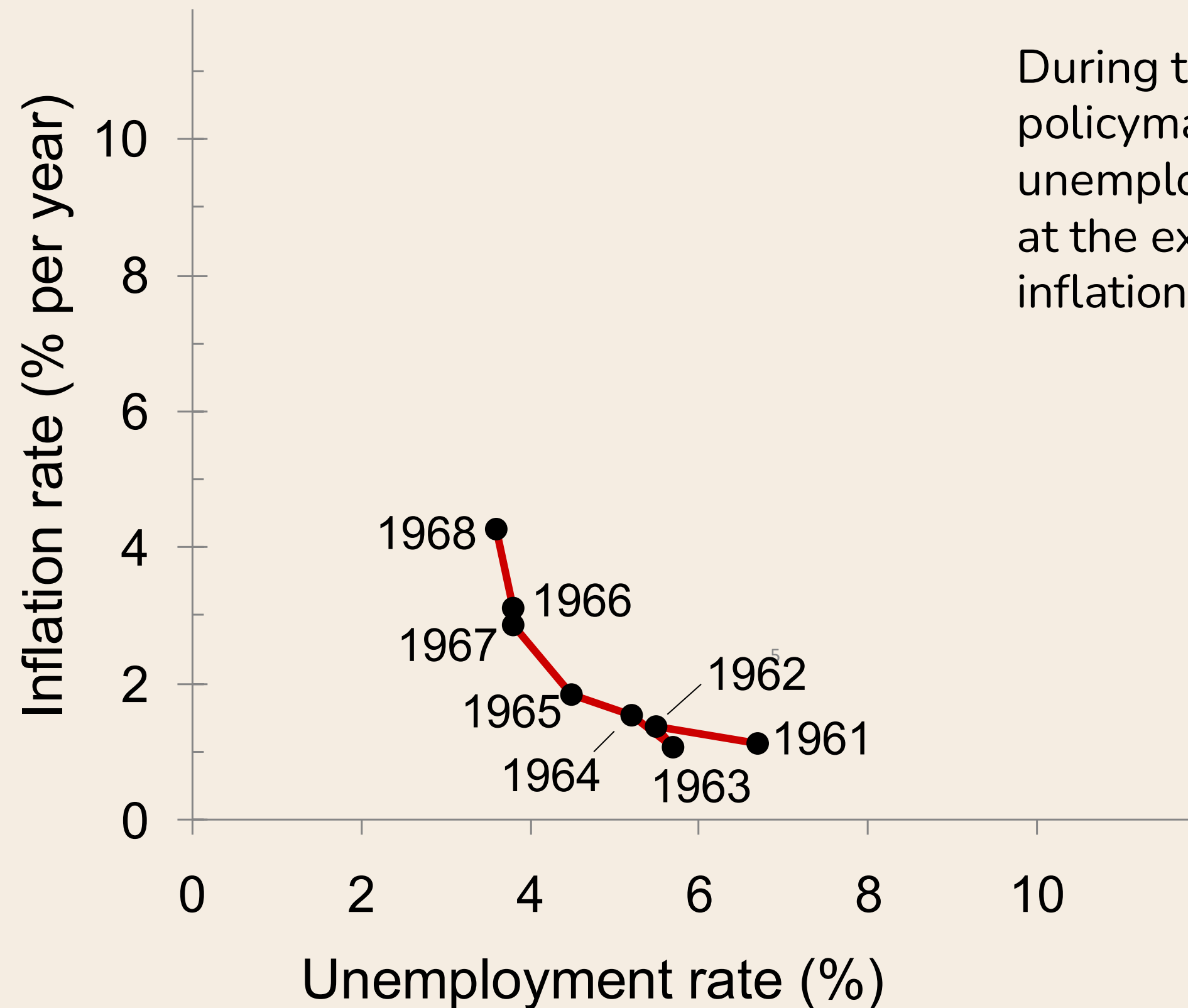
Point A: Low aggregate demand, low inflation, high u -rate

Point B: High aggregate demand, high inflation, low u -rate

The Phillips Curve: A Policy Menu?

- Since fiscal and monetary policy can shift the aggregate-demand curve,
 - They can move an economy along the **PC**
 - The **PC** offers policymakers a menu of combinations of inflation and unemployment:
 - Low unemployment with high inflation
 - Low inflation with high unemployment
 - Anything in between

Evidence for the Phillips Curve?



During the 1960s, U.S. policymakers opted for reducing unemployment at the expense of higher inflation

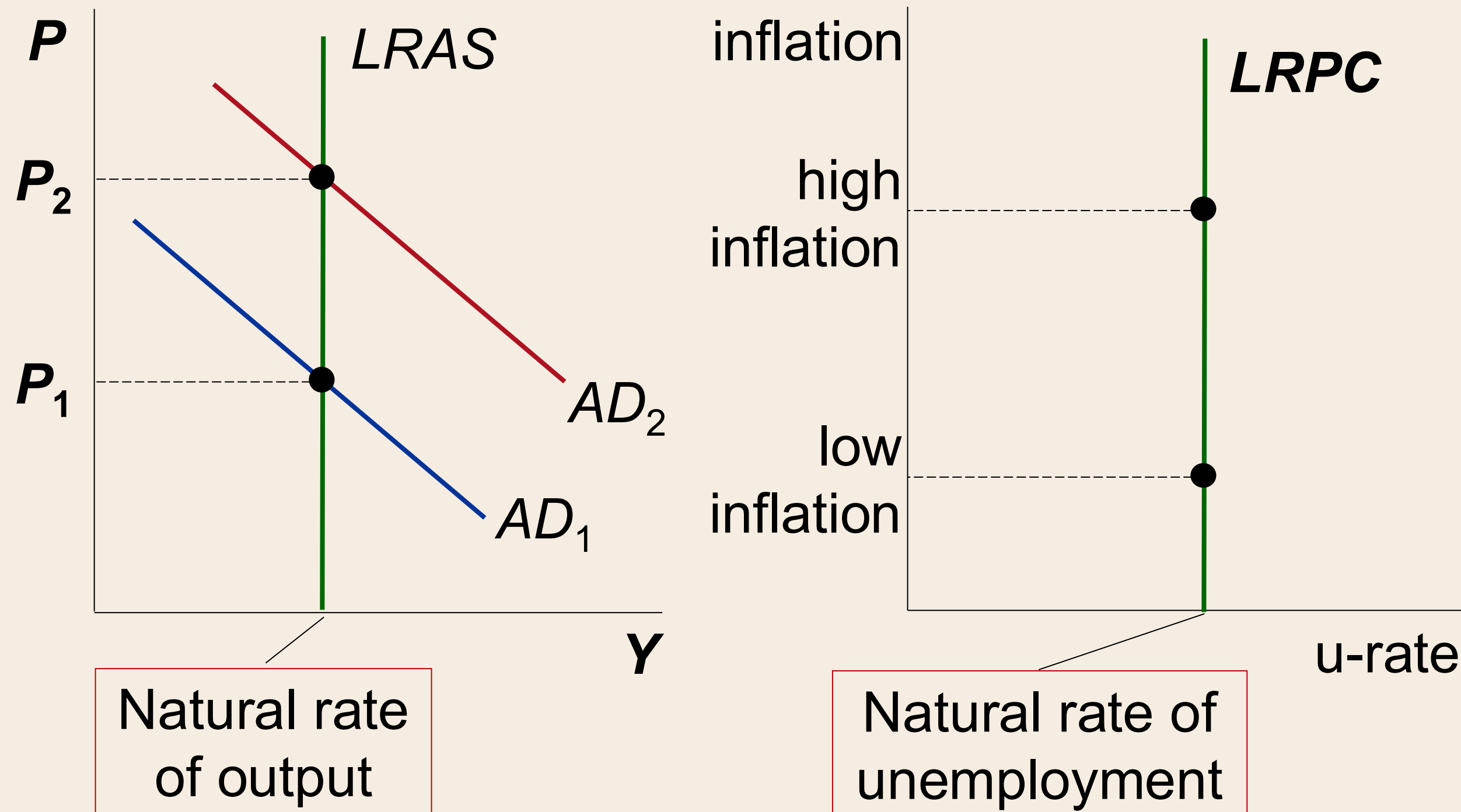
1960s: U.S. data supported the **PC**; Many believed the **PC** was stable and reliable

The Long-Run Phillips Curve

- 1968, Milton Friedman, Edmund Phelps:
 - Argued that the tradeoff was temporary
 - Based on the classical dichotomy and the vertical **LRAS** curve
- Natural-rate hypothesis:
 - The claim that unemployment eventually returns to its normal or “natural” rate, regardless of the inflation rate

The vertical long-run Phillips Curve, *LRPC*

In the long run, faster money growth only causes faster inflation.



The Meaning of “Natural”

- Natural rate of unemployment
 - Unemployment rate toward which the economy gravitates in the long run
 - Is not necessarily the socially desirable rate of unemployment
 - Is not constant over time
 - This unemployment is natural because it is beyond the influence of monetary policy

Reconciling Theory and Evidence

- Evidence (from 1960s):
 - **PC** slopes downward
- Theory (Friedman and Phelps):
 - **PC** is vertical in the long run.
- Friedman and Phelps, bridge the gap between theory and evidence
 - Introduced a new variable, expected inflation: a measure of how much people expect the price level to change

The Phillips Curve Equation

$$\text{u-rate} = \text{Natural u-rate} - a \left(\text{Actual inflation} - \text{Expected inflation} \right)$$

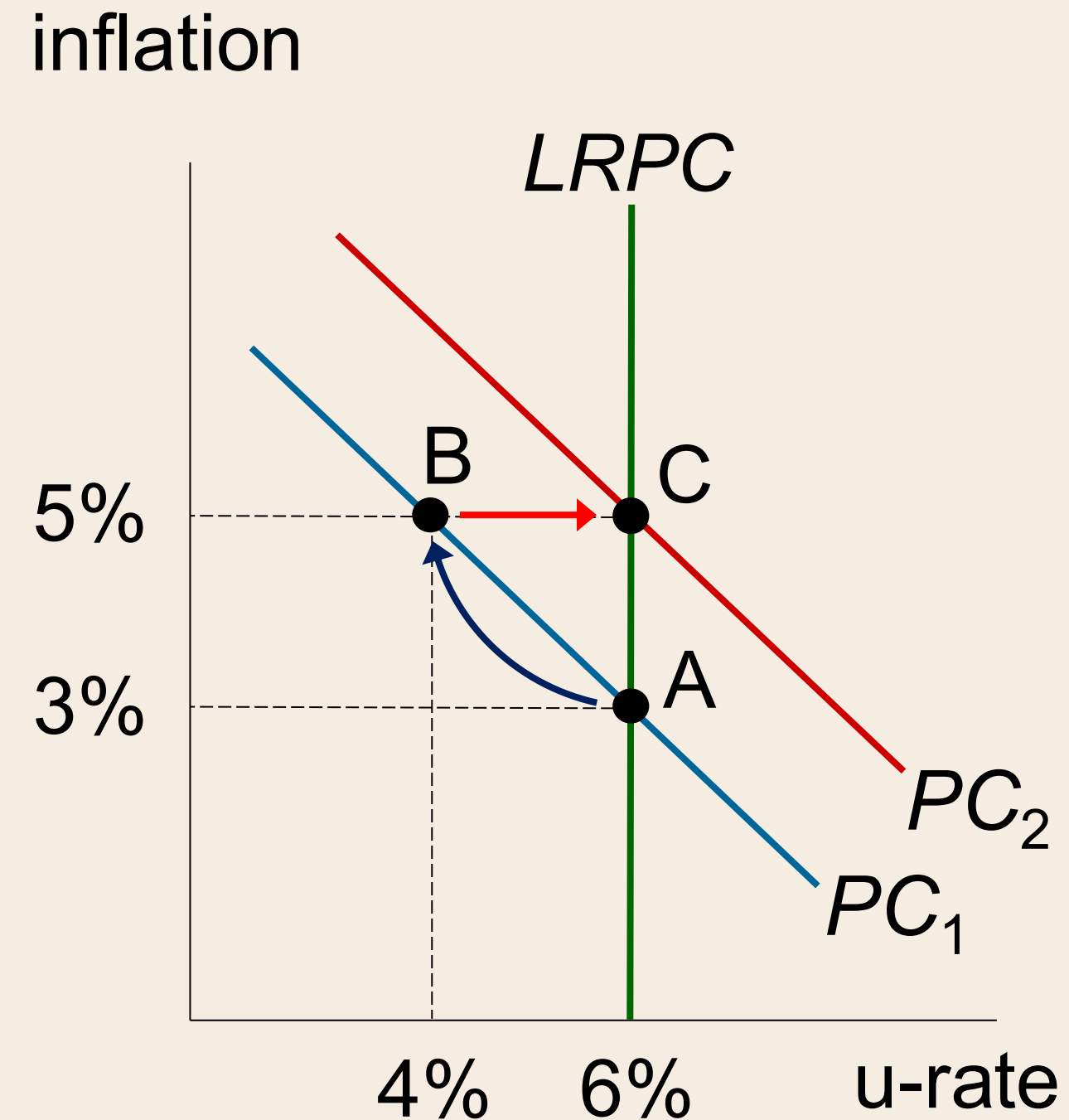
- Short run
 - The Fed can reduce u-rate below the natural u-rate by making inflation greater than expected.
- Long run
 - Expectations catch up to reality, u-rate goes back to natural u-rate whether inflation is high or low.

How expected inflation shifts the *PC*

Initially, expected & actual inflation = 3%,
unemployment = natural rate (6%).

Fed makes inflation
2% higher than expected,
u-rate falls to 4%.

In the long run, expected inflation
increases to 5%,
PC shifts upward,
unemployment returns to its natural rate.



Active Learning 1: A numerical example

Natural rate of unemployment = 5%

Expected inflation = 2%

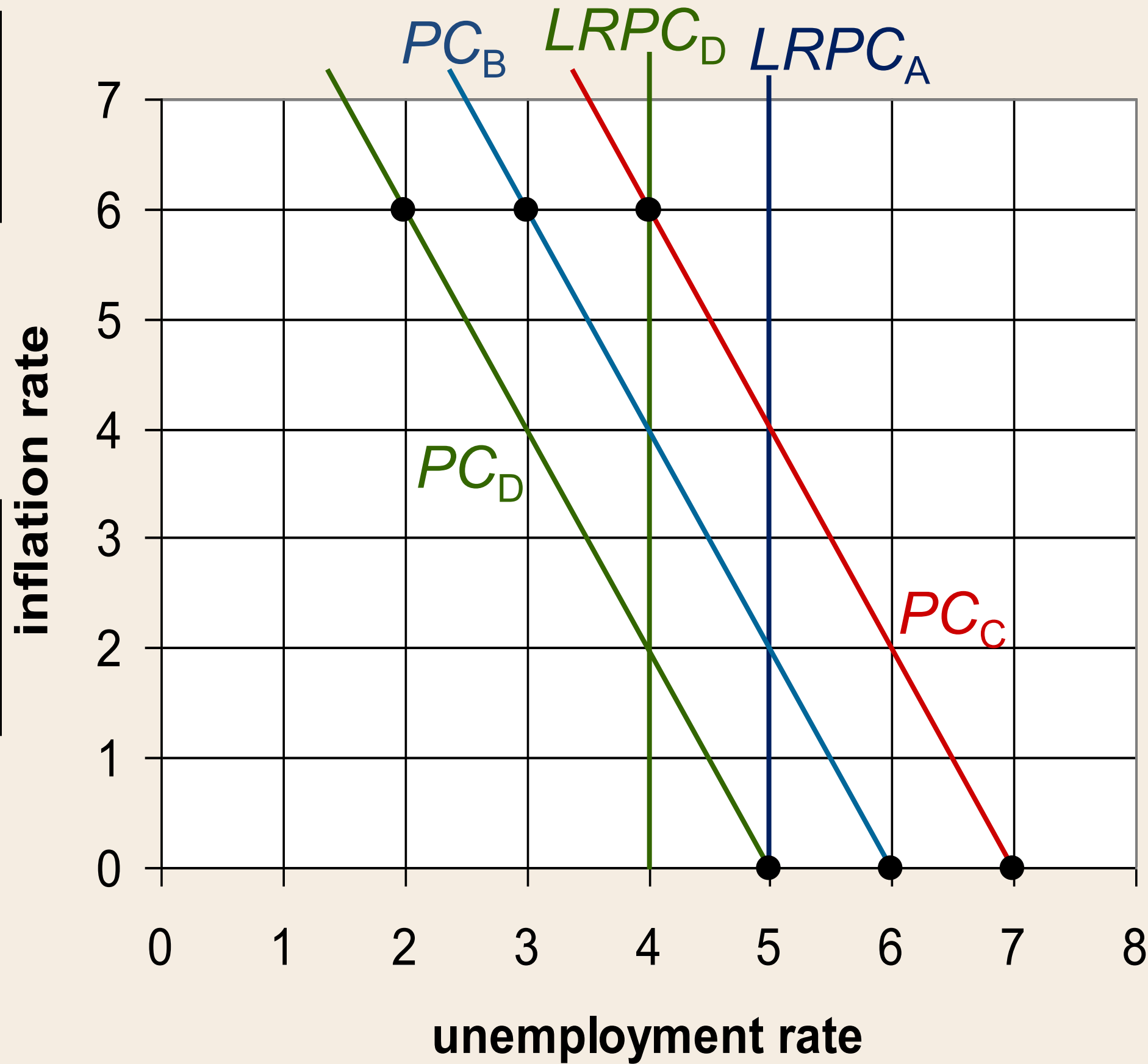
In **PC** equation, $a = 0.5$

- A. Plot the long-run Phillips curve.
- B. Find the u-rate for each of these values of actual inflation: 0%, 6%.
Sketch the short-run **PC**.
- C. Suppose expected inflation rises to 4%.
Repeat part B.
- D. Instead, suppose the natural rate falls to 4%. Draw the new long-run Phillips curve,
then repeat part B.

Active Learning 1: Answers

An increase in expected inflation shifts **PC** to the right.

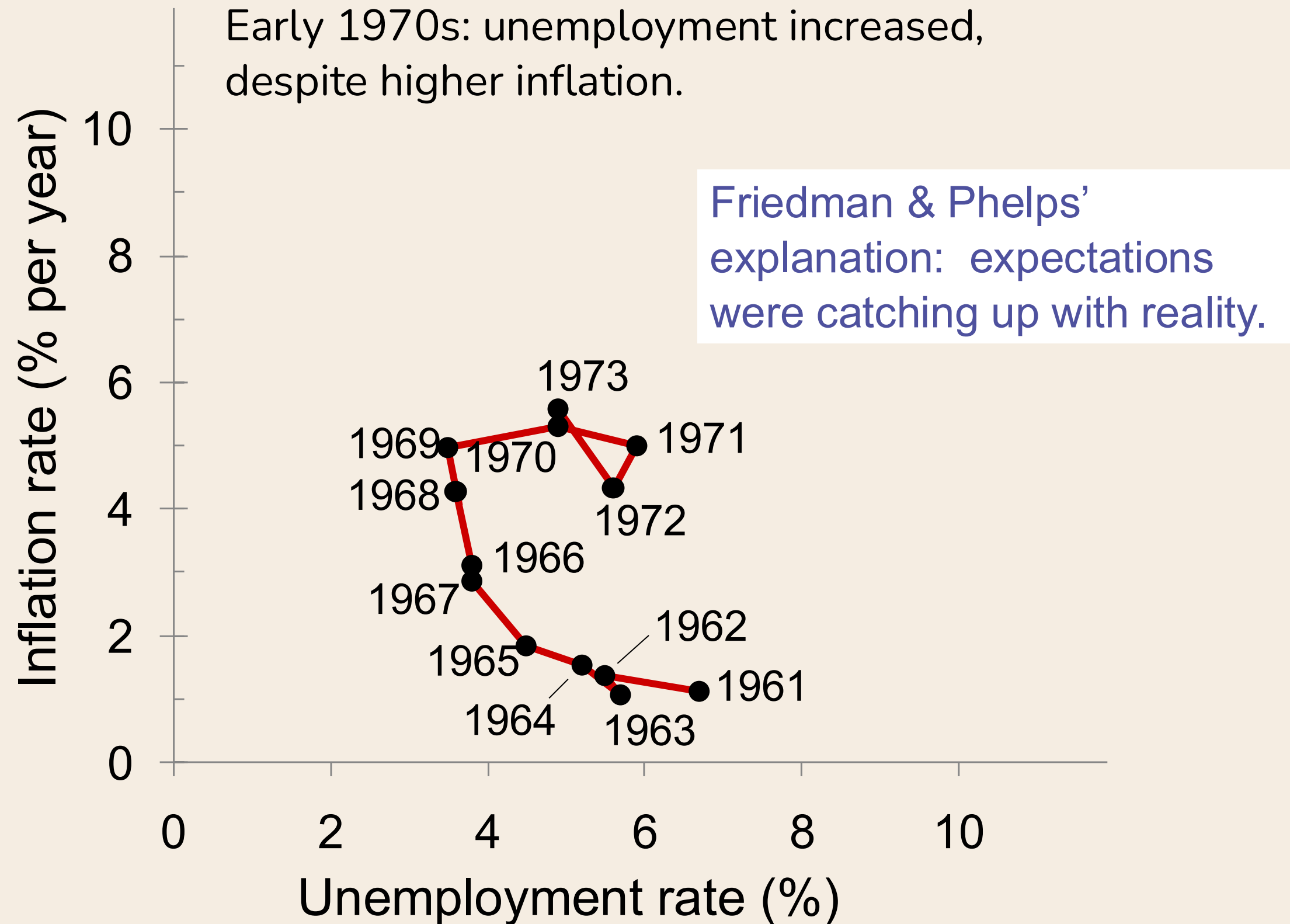
A fall in the natural rate shifts both curves to the left.



The Natural Experiment

- The natural-rate hypothesis
 - Unemployment eventually returns to its natural rate regardless of the rate of inflation
 - Predicted by Friedman and Phelps in 1968
 - Invertedly tested by policymakers in late 1960s, early 1970s

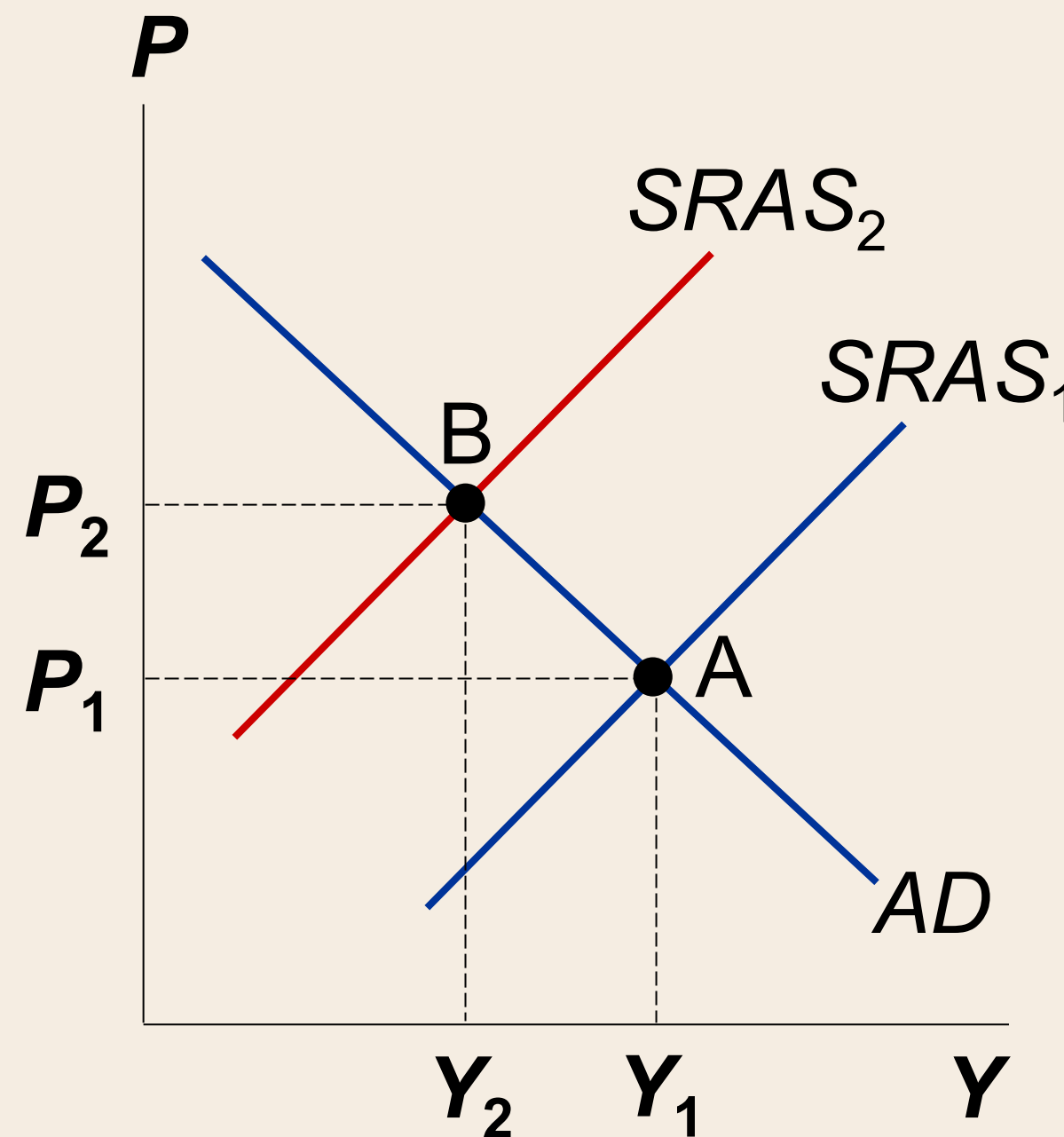
The breakdown of the Phillips Curve



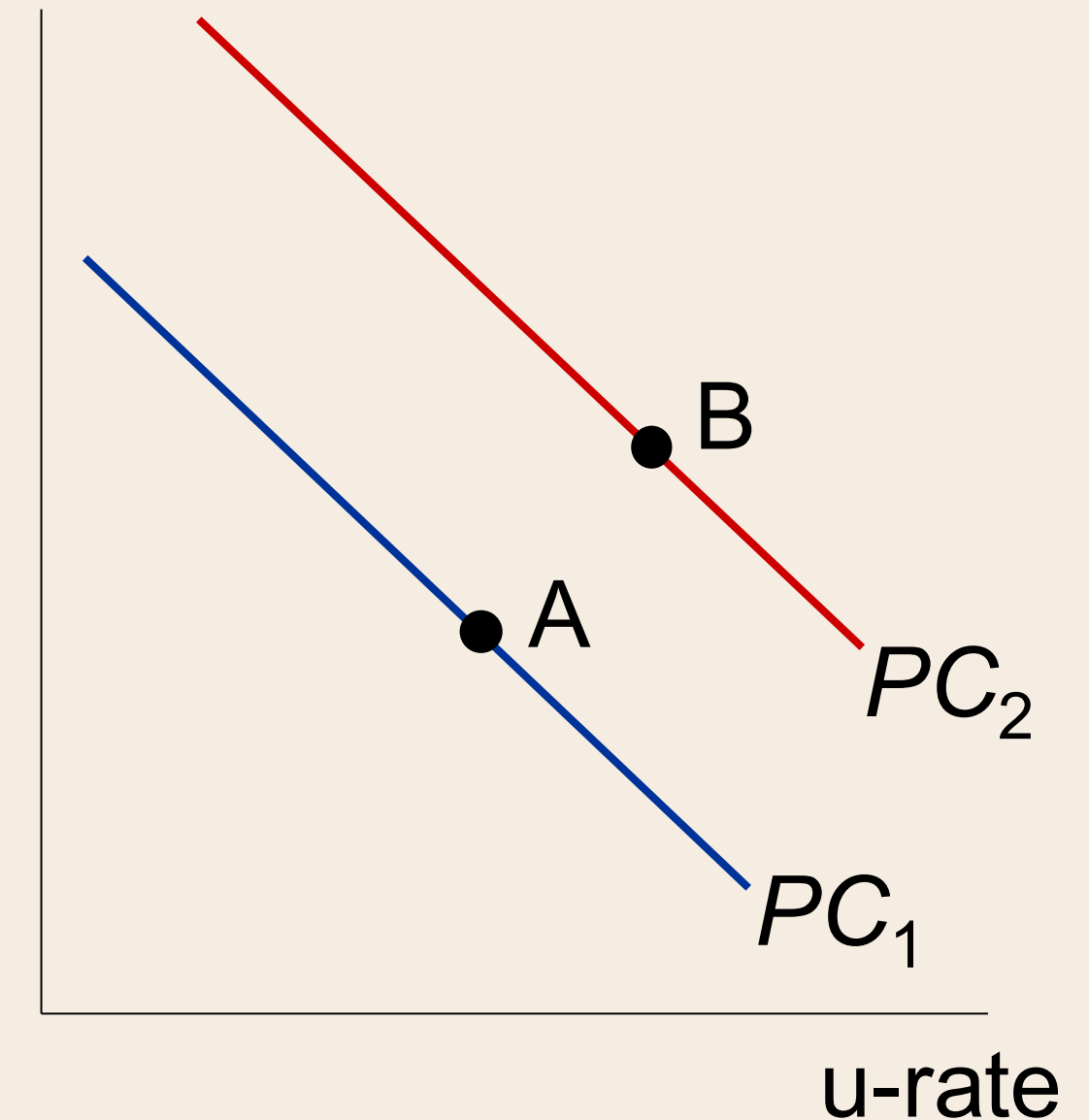
Supply Shocks and How an adverse supply shock shifts the *PC*

- Supply shock:
 - An event that directly alters firms' costs and prices
 - Shifting the **AS** and **PC** curves
 - Example: large increase in oil prices
 - 1974 and 1979, OPEC restricted the supply of oil: higher oil prices

SRAS shifts left, prices rise, output & employment fall.



inflation



Inflation & u-rate both increase as **PC** shifts upward.

The Cost of Reducing Inflation

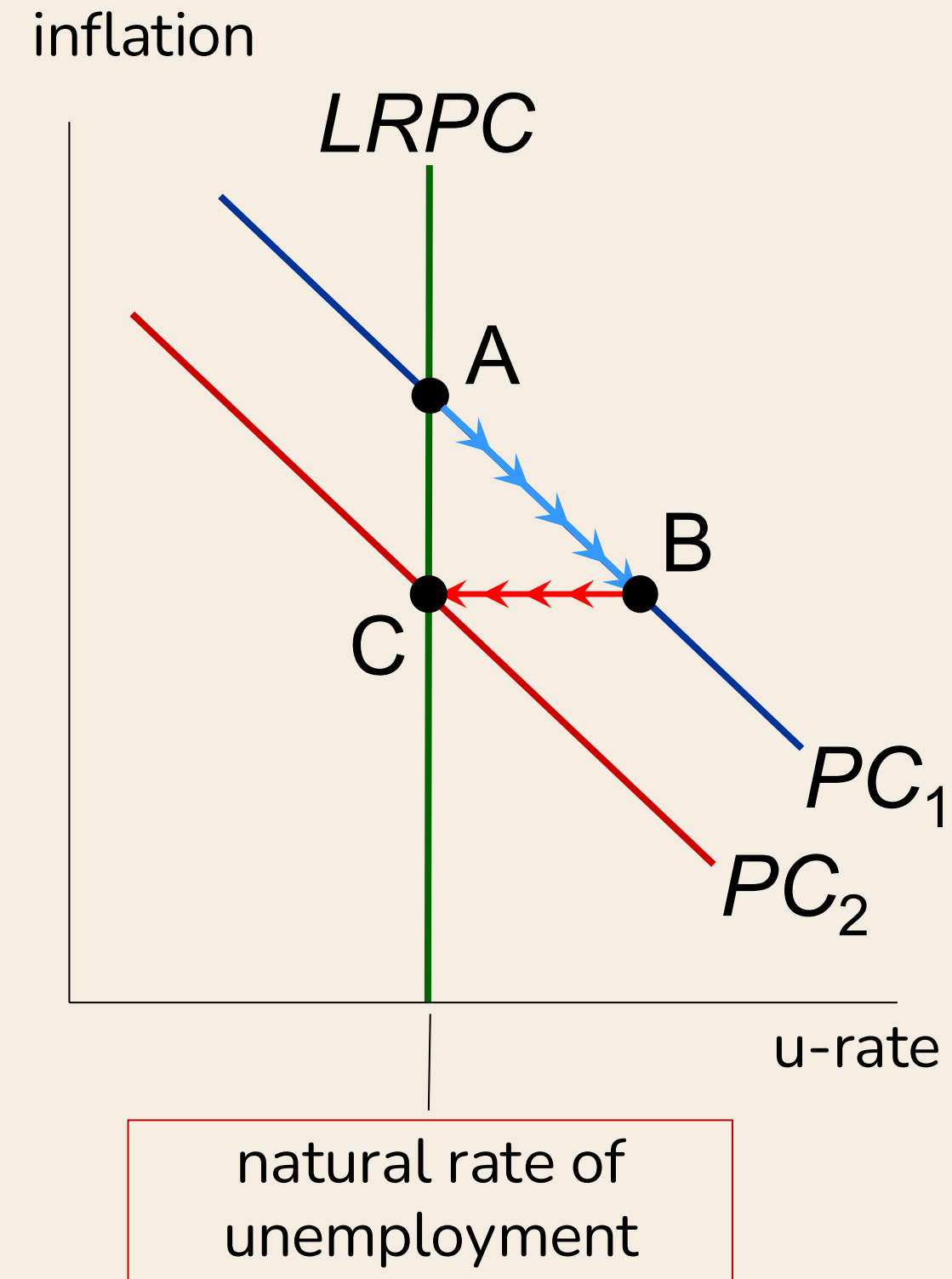
- Disinflation:
 - A reduction in the inflation rate
- To reduce inflation,
 - The Fed has to pursue contractionary monetary policy, which reduces **AD**
 - Short run: output falls and unemployment rises.
 - Long run: output & unemployment return to their natural rates.

Disinflationary monetary policy

Contractionary monetary policy moves economy from A to B.

Over time,
expected inflation falls,
PC shifts downward.

In the long run,
point C: the natural rate
of unemployment,
lower inflation.



Sacrifice Ratio

- Sacrifice ratio:
 - Percentage points of annual output lost per 1 percentage point reduction in inflation
 - Typical estimate: 5 (to reduce inflation rate 1%, must sacrifice 5% of a year's output)
 - Can spread cost over time: to reduce inflation by 6%, can either
 - sacrifice 30% of *GDP* for one year
 - sacrifice 10% of *GDP* for three years

Rational Expectations, Costless Disinflation?

- Rational expectations:
 - Theory that people optimally use all the knowledge they have, including information about government policies, when forecasting the future
 - Early proponents: Robert Lucas, Thomas Sargent, Robert Barro.
 - Implied that disinflation could be much less costly.

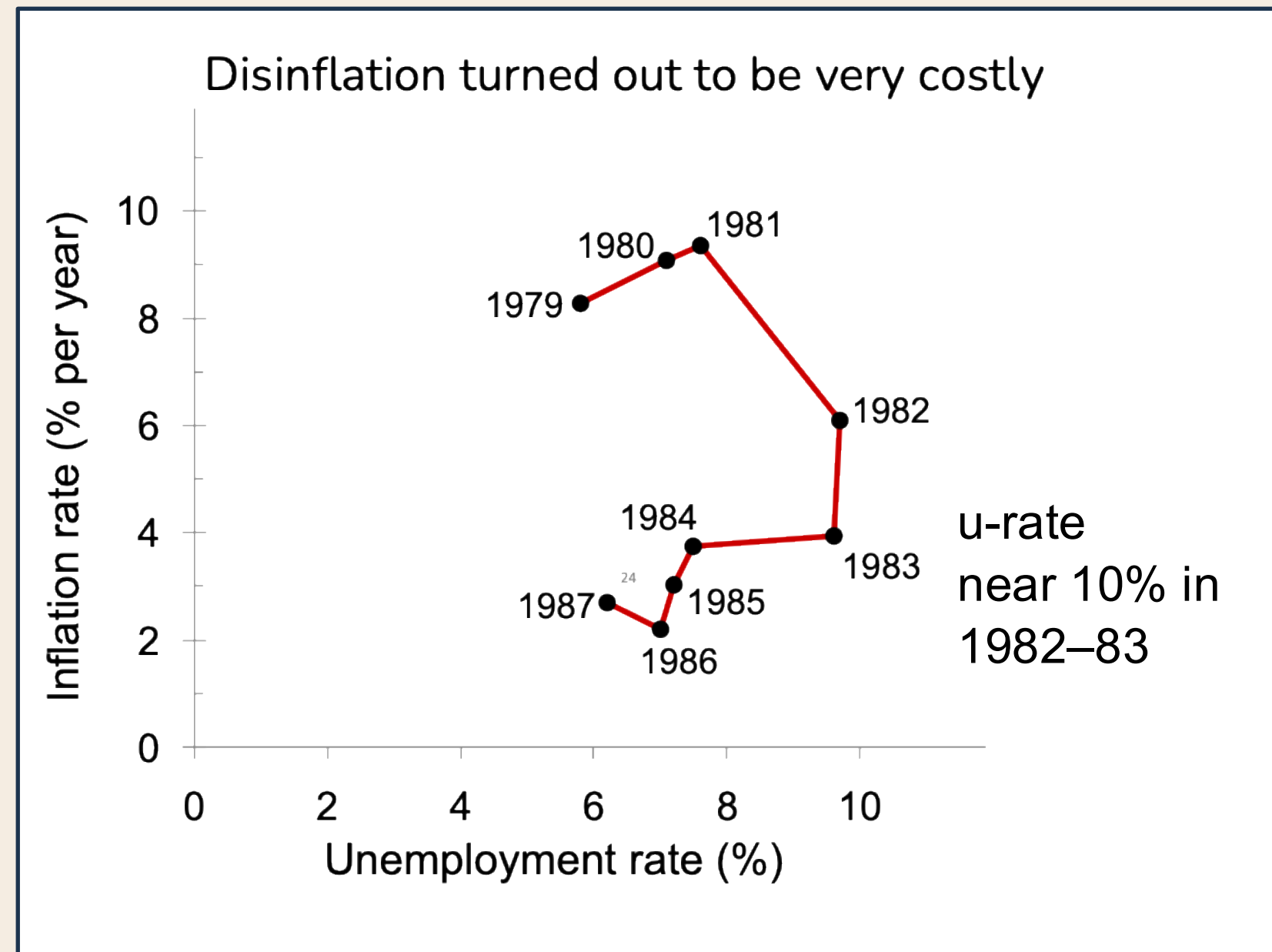
EXAMPLE 1: Rational expectations

Suppose the Fed convinces everyone it is committed to reducing inflation.

- Then, expected inflation falls, the short-run PC shifts downward.
- Result: disinflations can cause less unemployment than the traditional sacrifice ratio predicts.

The Volcker Disinflation

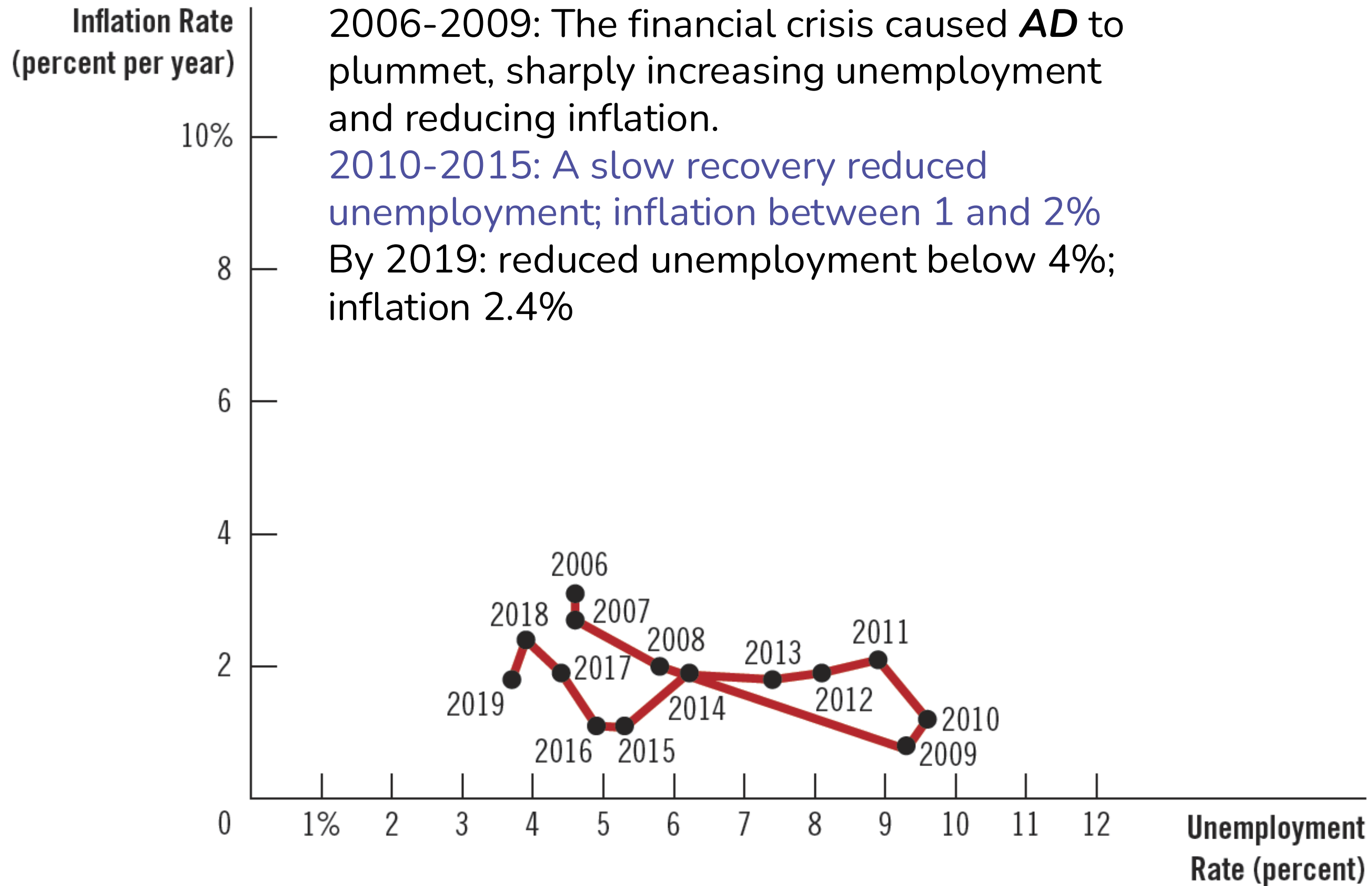
- Fed Chairman Paul Volcker
 - Appointed in late 1979 under high inflation & unemployment
 - Changed Fed policy to disinflation
- 1981–1984: Fiscal policy was expansionary
 - So, Fed policy had to be very contractionary to reduce inflation.
 - Success: Inflation fell from 10% to 4%,
but at the cost of high unemployment.



The Great Recession

- The early 2000s
 - Housing market boom turned to bust in 2006
 - Household wealth fell,
 - Millions of mortgage defaults and foreclosures
 - Heavy losses at financial institutions
- Result:
 - Sharp drop in aggregate demand, steep rise in unemployment
- 2007 to 2010, decline in AD
 - Raised unemployment (from below 5% to 10%)
 - Reduced the rate of inflation (from 3 to 1%)
- After 2010, slow recovery
 - Unemployment gradually declined
 - Rate of inflation remained between 1 and 2%
 - Expected inflation: steady at about 2%
 - Short-run Phillips curve relatively stable

Phillips Curve during and after the Great Recession–



The Pandemic

- 2020, coronavirus pandemic, sharp recession
 - Decline in **AD** : February to April 2020: u-rate jumped from 3.5 to 14.8%; CPI fell by 1% (brief period of deflation)
 - **AD** recovered quickly
 - Relaxation of restrictions
 - Expansionary monetary and fiscal policies
 - Consistently adverse effect on **AS**
 - Early: many businesses closed
 - After restrictions relaxed: people reluctant to return to work
 - Disrupted global supply chains, Further contracted AS
- **By the end of 2021**
 - U-rate was back below 5% (3.5% by July 2022)
 - Increased inflation
 - Adverse supply shocks and the expansion in aggregate demand from monetary and fiscal policy increased inflation.
 - 7.5% in January 2022
- Transitory inflation?

Conclusion

- Theories in this chapter teach us that inflation and unemployment are:
 - Unrelated in the long run
 - Negatively related in the short run
 - Affected by expectations, which play an important role in the economy's adjustment from the short-run to the long run

CHAPTER IN A NUTSHELL

- The Phillips curve (**PC**): negative relationship between inflation and unemployment.
 - By **expanding AD**: choose a point on **PC** with higher inflation and lower unemployment.
 - By **contracting AD**: choose a point on **PC** with lower inflation and higher unemployment.
 - This trade-off holds only in **the short run**.
- In the long run, expected inflation adjusts to changes in actual inflation, the short-run **PC** shifts: the long-run **PC** is vertical at the natural rate of unemployment.
- The short-run Phillips curve also shifts because of shocks to aggregate supply.
 - An **adverse supply shock** (increase in world oil prices), gives policy makers a less favorable trade-off between inflation and unemployment.
 - After an adverse supply shock, policymakers have to accept a higher rate of inflation for any given rate of unemployment or a higher rate of unemployment for any given rate of inflation.
- When **the Fed contracts growth in the money supply** to reduce inflation, it moves the economy along the short-run Phillips curve, resulting in temporarily high unemployment. The cost of disinflation depends on how quickly expectations of inflation fall.
- Some economists argue that a credible commitment to low inflation can reduce the cost of disinflation by inducing a quick adjustment of expectations.

Latihan Soal

Apa yang terjadi pada kurva Phillips jangka pendek (SRPC) ketika bank sentral meningkatkan jumlah uang beredar?

- a. Kenaikan jumlah uang beredar meningkatkan permintaan agregat, harga turun, dan ekonomi bergerak sepanjang kurva Phillips ke titik inflasi yang lebih tinggi dan tingkat pengangguran yang lebih rendah.
- b. Kenaikan jumlah uang beredar meningkatkan permintaan agregat, harga meningkat, dan ekonomi bergerak sepanjang kurva Phillips ke titik inflasi yang lebih tinggi dan tingkat pengangguran yang lebih rendah.
- c. Kenaikan jumlah uang beredar meningkatkan penawaran agregat, harga naik, dan ekonomi bergerak sepanjang kurva Phillips ke titik inflasi yang lebih rendah dan tingkat pengangguran yang lebih tinggi.
- d. Kenaikan jumlah uang beredar meningkatkan penawaran agregat, harga naik, ekonomi bergerak sepanjang kurva Phillips ke titik inflasi yang lebih tinggi dan tingkat pengangguran yang lebih rendah.

Ketika laju pertumbuhan uang beredar menurun, maka ekspektasi inflasi akan ... dan kurva Phillips jangka pendek (SRPC) akan bergeser

- a. menurun; ke bawah
- b. menurun; ke atas
- c. meningkat; ke bawah
- d. meningkat; ke atas

Kekeringan yang terjadi mengakibatkan harga pangan menjadi meningkat. Apa dampaknya terhadap tingkat inflasi dan pengangguran di jangka pendek?

- a. Kurva penawaran agregat bergeser ke kanan, dan kurva Phillips di jangka pendek akan geser ke kanan, karena biaya produksi meningkat. Kurva Phillips yang semakin tinggi artinya tingkat inflasi semakin tinggi pada tingkat pengangguran tertentu.
- b. Kurva penawaran agregat bergeser ke kiri, dan kurva Phillips di jangka pendek akan geser ke kiri, karena biaya produksi yang meningkat. Kurva Phillips yang semakin tinggi artinya tingkat inflasi semakin tinggi pada tingkat pengangguran tertentu.
- c. Kurva penawaran agregat bergeser ke kiri dan kurva Phillips di jangka pendek akan geser ke kanan, karena biaya produksi yang meningkat. Kurva Phillips yang semakin tinggi artinya tingkat inflasi semakin tinggi pada tingkat pengangguran tertentu.
- d. Kurva penawaran agregat akan bergeser ke kanan dan kurva Phillips jangka pendek akan geser ke kiri, karena biaya produksi yang meningkat. Kurva Phillips yang semakin tinggi artinya tingkat inflasi semakin tinggi pada tingkat pengangguran tertentu.

Negara A memiliki *sacrifice ratio* sebesar 5, sedangkan Negara B memiliki *sacrifice ratio* sebesar 1. Manakah pernyataan di bawah ini yang benar?

- a. Negara A harus meningkatkan jumlah uang beredar 5 kali lebih banyak untuk mengurangi inflasi dibandingkan Negara B
- b. Negara A harus berinvestasi 5 kali lebih banyak untuk mencapai tingkat pertumbuhan yang sama dengan Negara B
- c. Untuk mengurangi tingkat inflasi yang sama, jumlah output yang berkurang di Negara A lebih rendah daripada di Negara B
- d. Turunnya lapangan pekerjaan di Negara A lebih besar untuk mengurangi tingkat inflasi yang sama.

Krisis Timur Tengah baru-baru ini telah menciptakan tekanan pada logistik. Krisis ini telah berdampak pada aktivitas perdagangan di wilayah tersebut. Sebuah studi dari lembaga ekonomi Jerman IfW Kiel menemukan bahwa jumlah kontainer yang melewati wilayah Laut Merah menurun sebesar 60% akibat krisis pada awal tahun 2024. Hal ini, ditambah dengan pengumuman pemotongan produksi minyak OPEC pada Maret 2024, telah mendorong harga minyak naik. Manakah dari berikut ini yang merupakan dampak langsung jangka pendek dari krisis tersebut?

- a. Permintaan agregat bergeser ke kiri, menggeser kurva Phillips jangka pendek ke kanan, meningkatkan inflasi dan pengangguran.
- b. Permintaan agregat bergeser ke kiri, menggeser kurva Phillips jangka pendek ke kanan, meningkatkan pengangguran tetapi mengurangi inflasi.
- c. Penawaran agregat bergeser ke kiri, menggeser kurva Phillips jangka pendek ke kanan, meningkatkan inflasi dan pengangguran.
- d. Penawaran agregat bergeser ke kiri, menggeser kurva Phillips jangka pendek ke kiri, meningkatkan pengangguran tetapi mengurangi inflasi.

Esai:

Jawablah pertanyaan-pertanyaan di bawah ini dengan menggunakan analisis verbal dengan menggunakan diagram yang relevan.

- a. Untuk mengantisipasi pelemahan perekonomian global, pemerintah mengimplemen-tasikan kebijakan fiskal ekspansif. Jelaskan dampak kebijakan fiskal ekspansif pemerintah terhadap perekonomian dalam jangka pendek menggunakan model AD- SRAS dan *Short Run Phillips Curve*. Identifikasi apa yang terjadi pada tingkat output, tingkat harga, laju inflasi, dan tingkat pengangguran.
- b. Saat otoritas moneter dalam sebuah perekonomian relatif kredibel dalam mengendalikan inflasi, *cost of disinflation* dalam perekonomian bisa relatif rendah sehingga kebijakan moneter kontraktif untuk menurunkan inflasi tidak menyebabkan kenaikan pengangguran berlangsung lama. Jelaskan secara lengkap menggunakan perangkat kurva AD-AS dan Phillips Curve dalam jangka pendek dan jangka panjang.

Perdana Menteri Brogentina sedang merancang sebuah kebijakan untuk menyeimbangkan tingkat inflasi dan tingkat pengangguran. Dari data BPS Brogentina, tingkat pengangguran natural dalam 20 tahun terakhir sebesar 6%, tingkat ekspektasi inflasi masyarakat yang disurvei di awal tahun 2023 adalah 5%, dan diketahui koefisien α pada persamaan kurva Phillips adalah 0.2.

- a. Gambarkan kurva Phillips jangka pendek dan jangka panjang pada tingkat inflasi 3% dan 10%.
- b. Gambarkan proses penyesuaian dari tingkat inflasi dan pengangguran jika pada akhir tahun 2023 ekspektasi inflasi meningkat menjadi 6%.
- c. Jika pada jangka panjang, pengangguran natural diproyeksikan menurun 4%, apa saja kebijakan yang bisa menggeser tingkat pengangguran lebih kecil pada jangka panjang?

